



COMPUTER VISION FUNDAMENTALS

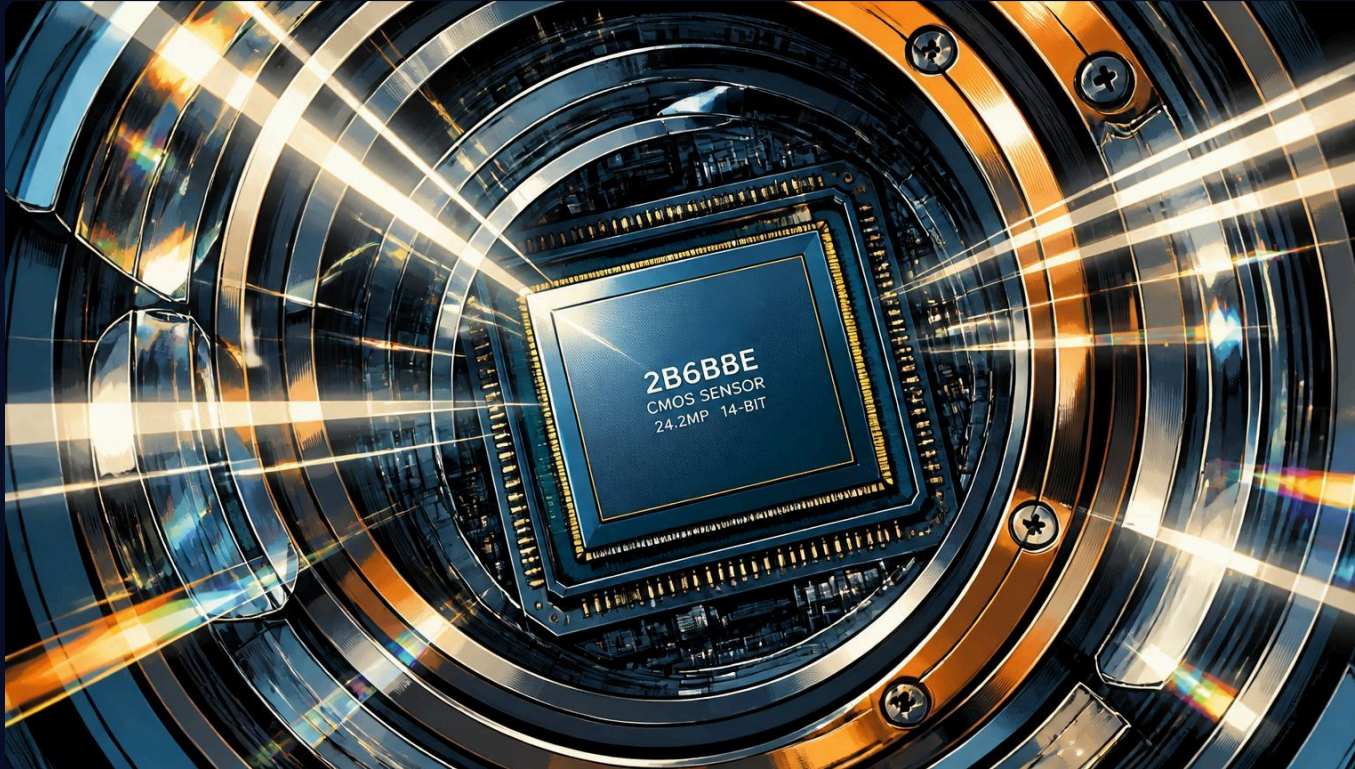
From Pixels to Intelligence

Computer Vision enables machines to capture, process, and understand visual data – combining digital image processing, machine learning, and pattern recognition to make intelligent decisions across healthcare, autonomous vehicles, agriculture, surveillance, robotics, and retail.

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Digital Image Fundamentals & Image Acquisition



What Is a Digital Image?

A digital image is a grid of **pixels**, each storing brightness or color. Key properties include **spatial resolution** (e.g., 1920×1080), **intensity resolution** (8-bit = 256 levels), and **grayscale** intensity values.

Image Acquisition Chain

01

Real-World Scene

02

Camera & Lens

03

CCD / CMOS Sensor

04

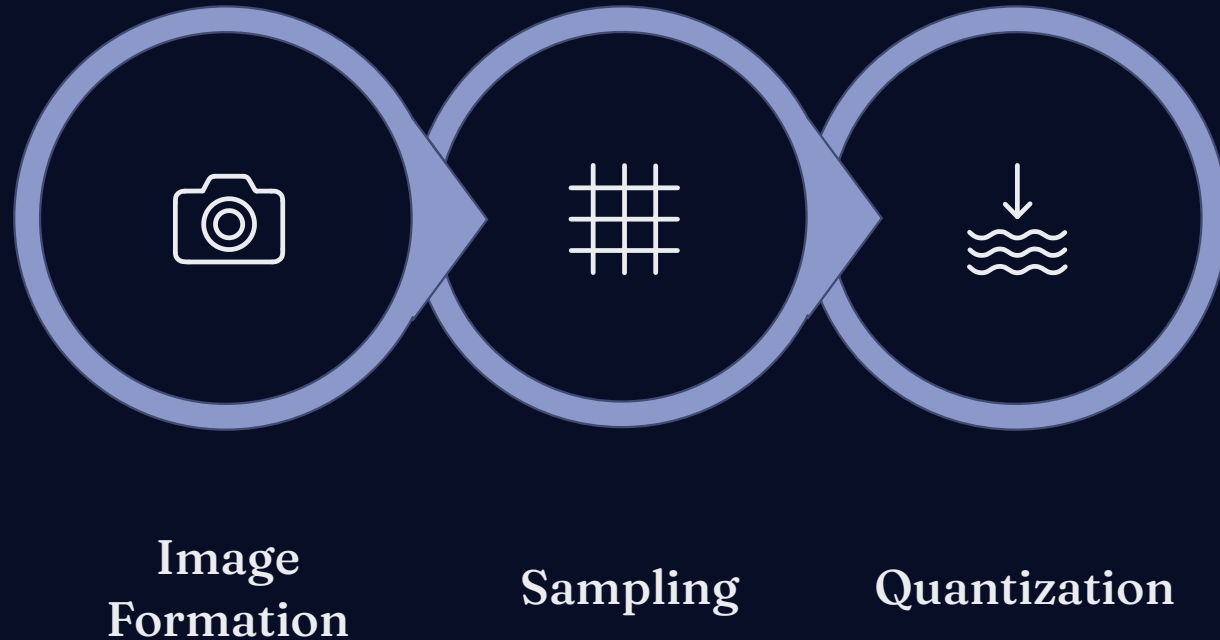
Analog-to-Digital Converter

05

Digital Image

Image Formation, Sampling & Quantization

Three sequential steps transform a continuous real-world scene into a usable digital image.

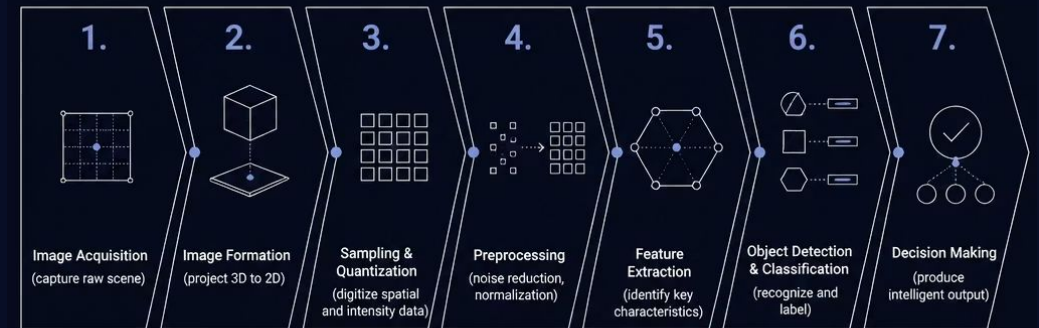


Each stage introduces trade-offs: **image formation** handles projection and lens effects, **sampling** controls spatial detail, and **quantization** determines tonal fidelity. Together they define the quality ceiling for all downstream vision tasks.

The Complete Computer Vision Pipeline

Every stage is **interconnected** – weaknesses at any point cascade through the entire system. High-quality acquisition and accurate formation ensure reliable input; proper sampling and quantization preserve detail; preprocessing and feature extraction drive recognition accuracy.

📄 Better image quality at the front end directly improves downstream performance – from feature extraction to final decision-making.



Real-World Engineering Applications



Healthcare

Tumor detection from MRI/CT scans and disease diagnosis from X-rays.



Autonomous Vehicles

Lane detection, traffic sign recognition, and pedestrian detection.



Agriculture

Crop disease detection and smart irrigation monitoring.



Surveillance

Face recognition, intrusion detection, and smart city monitoring.



Retail

Automated billing, customer behavior analysis, and inventory management.

 **Common Challenges:** Noise, poor illumination, motion blur, lens distortion, and low resolution.



Key Takeaways

Interconnected Pipeline

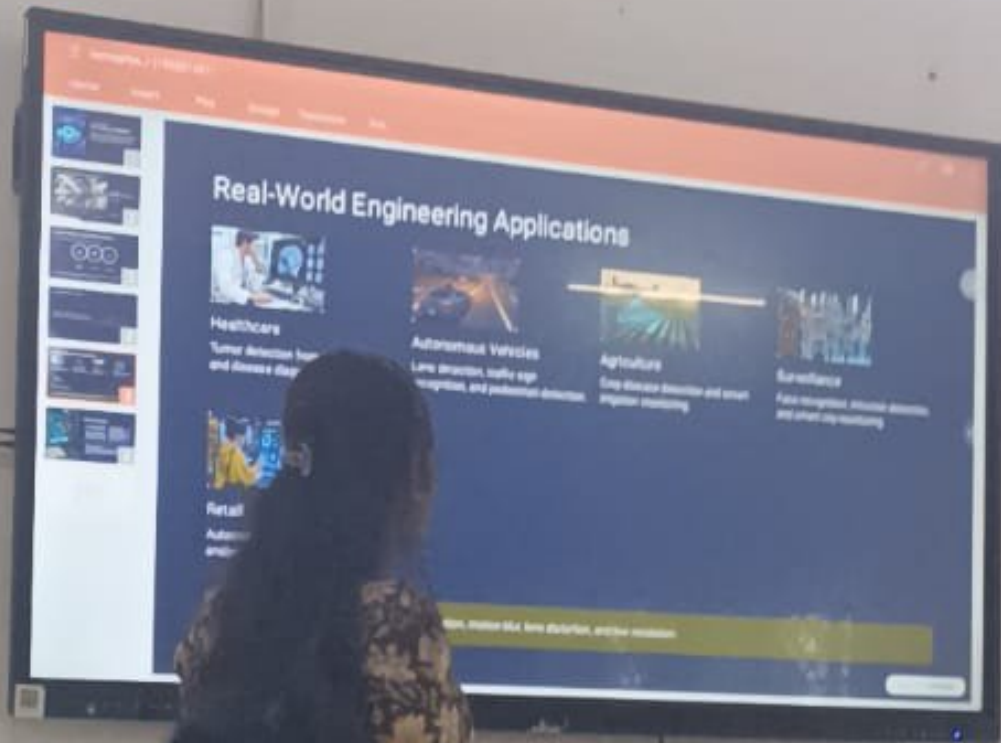
Every stage – from acquisition to decision-making – depends on the quality of those before it.

Quality Matters

High-quality acquisition, accurate formation, and proper sampling directly improve recognition accuracy.

Engineering Impact

Computer Vision is transforming healthcare, transportation, agriculture, and beyond through intelligent automation.



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CO1 - AT3 - TECHNICAL PRESENTATION

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Course: Computer Vision.

Interconnection of Digital Image Fundamental,
Image Acquisition, Image Formation, Sampling, Quantization
and Computer Vision Application.

Technical Content Accuracy (30)	Problem Identification (25)	use of Tools (20)	Communication (15)	Proficiency - skill (10)	Total (100)
28	25	20	15	10	98

[Signature]